

Exploratory Data Analysis

This phase is the investigation phase of data analysis which helps the data analyst to avoid making wrong calculations from messy or misunderstood data. The main purpose of this phase is to prepare the data for modelling or reporting which involves understanding the structure of the data, identifying outliers, trends, and pattern, checking assumptions, finding relationships between variables, etc.

Overview

This table below shows the overall view of the basic metrics

Basic metrics	Values
Total Orders	1200
Total Revenue	\$ 1,264,762
Avg Order value	1054
Missing data	309
Missing data (%)	25.8%
Cancelled orders	20.8%
Returned orders	20.6%
Sum of order cancelled and returned	41.4%

Basic Metrics	Min	Mean	Median	Max
Quantity	1	3	3	5
Unit price	\$ 11	\$ 356	\$ 364	\$ 700
Items in cart	1	5	5	10
Revenue	\$ 11	\$ 1,054	\$ 824	\$ 3,456

Note: The missing data has been fixed with "No coupon"

Outlier Checks

Outlier check	
First Quartile	411
Third Quartile	1578
Interquartile range	1168
Lower Bound	-1341
Upper Bound	3330

Outliers			
Order ID	Date	Product	Total Revenue
ORD200326	2024-07-01	Laptop	\$ 3,352
ORD200469	2023-11-26	Chair	\$ 3,385
ORD200328	2023-02-28	Tablet	\$ 3,370
ORD200632	2023-05-02	Laptop	\$ 3,391
ORD201065	2023-10-30	Printer	\$ 3,334
ORD200107	2023-03-27	Printer	\$ 3,354
ORD201122	2023-06-07	Monitor	\$ 3,391
ORD200789	2023-08-17	Tablet	\$ 3,456

The data above contains orders that can affect data accuracy. Since there is no equal distribution of prices for each good, the outliers were not removed.

Revenue performance

Year	Revenue (MoM Growth)
2023	
2024	-13.1%
2025	-51.7%
Grand Total	

Total Revenue	Year				
Payment Method	2023	2024	2025		Grand Total
Cash	10.83%		6.69%	3.02%	20.54%
Credit Card	8.98%		8.46%	3.42%	20.86%
Debit Card	7.91%		7.37%	3.09%	18.37%
Gift Card	8.39%		7.89%	3.20%	19.48%
Online	7.58%		7.56%	5.61%	20.75%
Grand Total	43.70%		37.97%	18.33%	100.00%

There revenue generated by the firm keeps decreasing over the years. A conclusion cannot be made that the firm's performance keeps depreciating since the prices of goods for each product ranges from \$11 to \$700 which implies that there are no fixed price for the goods.

The payment method are evenly distributed with Debit card and Gift card performing below average.

Product performance

Row Labels	Orders	Total Revenue		Average order value		Revenue share
Chair	178	\$	195,620	\$	1,099	15.5%
Desk	170	\$	167,460	\$	985	13.2%
Laptop	173	\$	192,127	\$	1,111	15.2%
Monitor	163	\$	175,651	\$	1,078	13.9%
Phone	156	\$	151,722	\$	973	12.0%
Printer	181	\$	195,613	\$	1,081	15.5%
Tablet	179	\$	186,569	\$	1,042	14.8%
Grand Total	1200	\$	1,264,762	\$	1,054	100.0%

Products are tightly clustered in revenue (Chair leads at \$195.6k, Phone lowest at \$151.7k).

Laptops have the highest average order value (\$1,111) despite fewer orders.

Order Status Performance

Order status	Order status (%)
Cancelled	20.8%
Delivered	19.3%
Pending	19.8%
Returned	20.6%
Shipped	19.6%
Grand Total	100.0%

Only 19.3% of the orders are delivered.

41.4% of the total orders were either cancelled or returned

This needs to be looked into immediately to understand the delivery issues and whether the products truly meets customer needs.

Referral Source Performance

Referral source	No. Of Orders	Average order value
Email	250	1,047
Facebook	228	1,098
Google	241	1,039
Instagram	259	1,063
Referral	222	1,022
Grand Total	1200	1053.9683

Instagram drives the most revenue though Facebook leads in average order value. Referral source has the lowest volume and lowest average order value.

All channels are within 17% of each other in total revenue, suggesting a well-diversified acquisition strategy.